



AB1620 Tool SDK Introduction

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Agenda

- ❑ **AB1620 Tool List**
- ❑ **1620 MP Library**
 - **Production Flow**
 - **1620 FW Download**
 - **1620 Crystal Trim**
 - **1620 RF test**
 - **1620 Wired Pairing**
- ❑ **1620 DFU/FOTA**
- ❑ **1620 HID : Windows UT App**
- ❑ **Tool SDK Debug**

AB1620 Tool List

- ❑ SDK folder : [airoha/mcu/tools/pc_tools/IOT_Flash_Tool](#)
 - Airoha_Flash_Tool_v1.1.20.0.7z
- ❑ SDK folder : [airoha/mcu/tools/pc_tools/atk/ab162x](#)

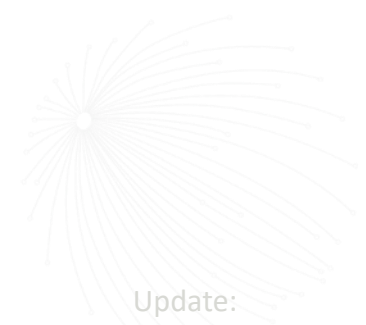
Tool	Description	SDK support
AB1620_Airoha_Tool_Kit	Logging/Lab test/Config/FOTA package	
AB1620_MP_Tool	Mass production tool	
AB1620_MPLibrary_SDK	MP Library SDK + sample code	Library + sample code
Airoha_BTD_USB_DFU	USB-HID DFU	Full sample code
Airoha_Bootloader_DFU	Bootloader USB DFU	Full sample code
Airoha_Factory_Data_Tool	Smart Home Matter Factory Data tool	Full sample code
Airoha_BTD_Wired_Pairing_Tool	HID Wired Pairing Tool	Library + sample code
Airoha_BTD_Android_SDK	Android FOTA App	Library + sample code
Airoha_BTD_iOS_SDK	iOS FOTA App	Library + sample code
Airoha_Windows_BTD_UT_APP	HID Windows APP	Library + sample code



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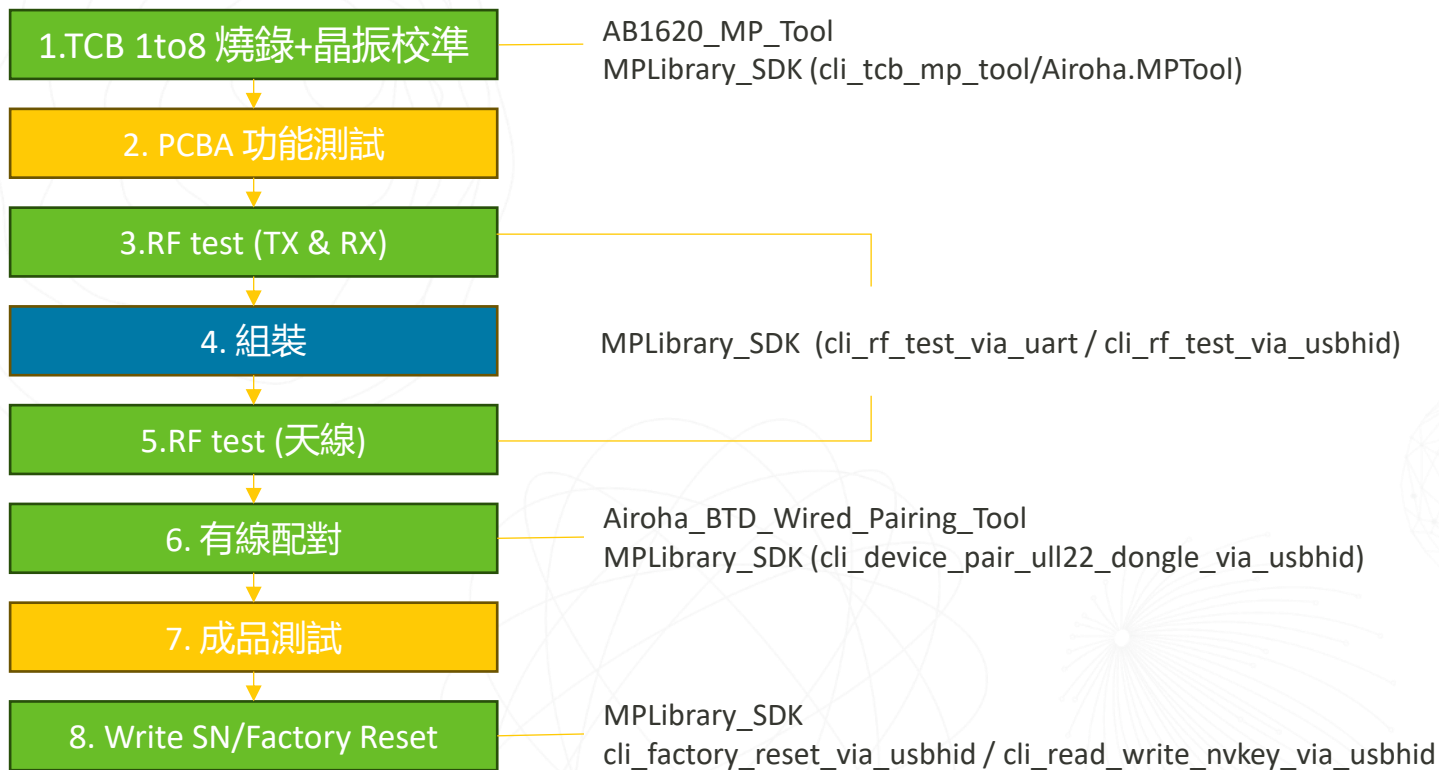


1620 MP Library



Update:

Production Flow



1620 FW Download

Tool	EXE or project	Description
Airoha_Flash_Tool	AIROHA_Flash_Tool.exe	UI tool
	CODA.exe	command-line tool read/write e-Fuse & OTP
AB1620_MP_Tool	Airoha.MPTool.exe	UI tool : using TCB 1to8 download
AB1620_MPLibrary_SDK	cli_tcb_mp_tool	Sample code using TCB (command-line)
	cli_write_flash_via_uart	UART download sample code
	Airoha.MPTool	MP tool UI sample code (TCB 1to8)

1620 Crystal Trim

Test HW	Tool	Function or API
Airoha TCB	MP tool	Crystal trim after FW download (+-2ppm)
RF instrument	AB1620_MPLibrary_SDK (cli_rf_test_via_uart)	SetCrystalCap (cap 0~511, initial : 111) TxDcTone ReadCrystalCap32M (flash) WriteCrystalCap32M (flash)

1620 RF test

Tool		Function
Lab test tool		LE single tone/Tx/Rx 802.15.4 single tone/Tx/Rx
AB1620_MPLibrary_SDK	cli_rf_test_via_uart	LE single tone/Tx/Rx
	cli_rf_test_via_usbhid	LE single tone/Tx/Rx
	cli_rf_test_802154_via_uart	802.15.4 single tone/Tx/Rx

1620 RF test – 2 devices Tx Rx test

- ❑ Using 1620 as golden sample to test DUT
- ❑ AB1620_MPLibrary_SDK (cli_rf_test_via_usbhid)

Create 2 handle

```
void DutTesting_2Device_TxRx()
{
    int handle1 = GetIf()->getHandler();
    int handle2 = GetIf()->getHandler();
    try
    {
        RF_Test_Process_2Device_TxRx(handle1, handle2);

        DisconnectUsbHid(handle1);
        DisconnectUsbHid(handle2);
    }
    catch (...)
    {
    }

    Sleep(1000);
    GetIf()->destroyHandler(handle1);
    GetIf()->destroyHandler(handle2);
}
```

Dongle
Mouse

Test 3 channels in 6 seconds

```
int test_freq[] = {2402, 2444, 2480};
printf("=====Test Dongle TX & Mouse RX=====\\n");
for (int i=0;i<3;++i)
{
    frequency = test_freq[i];
    printf("frequency=%d\\n", frequency);

    //DUT1 Tx start
    ret = LeTx(handle1, "1M");
    if (ret != 0)
        return;

    //DUT2 Rx start
    ret = LeRxStart(handle2, "1M");
    if (ret != 0)
        return;

    //DUT2 Rx stop , report : PER, RSSI
    ret = LeRxStop(handle2);
    if (ret != 0)
        return;

    //DUT1 Tx stop
    ret = LeTxStop(handle1);
    if (ret != 0)
        return;
}
```

Dongle Tx
Mouse Rx
Mouse Rx Stop (PER, RSSI)
Dongle Tx stop

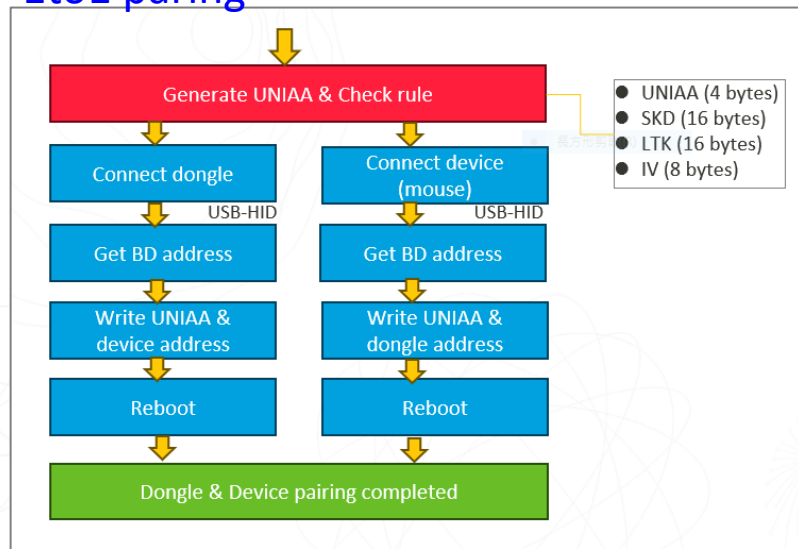
1620 HID wired pairing

Tool

Airoha_BT_D_Wired_Pairing_Tool

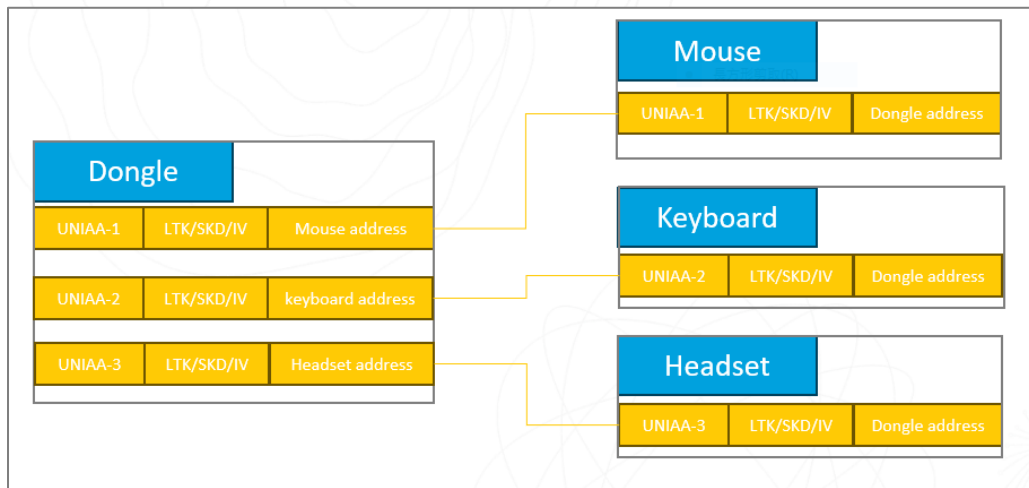
AB1620_MPLibrary_SDK
(cli_device_pair_ull22_dongle_via_usbhid)

1to1 pairing

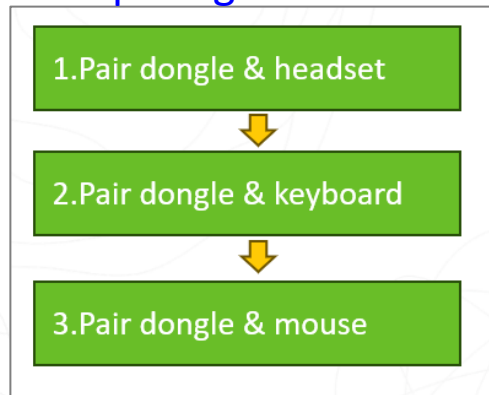


1620 HID wired pairing : 1 to N

❑ Dongle keep multi-pairing info



1to3 paring



1620 HID wired pairing : UNIAA rule

❑ 1 to 1 pairing

- Do not start with 0xFF 0xFE.
- Can not be all 0x00.
- Can not be all 0x01.
- Can not be 0xD6 BE 89 8E.
- No more than 6 bit consecutive zero or ones.
- All 4 octets are not equal.
- Limit to 24 transitions.
- Minimum of two transitions in the most significant six bits.

❑ 1 to N pairing

- All UNIAA in the dongle differ by at least 2 bits

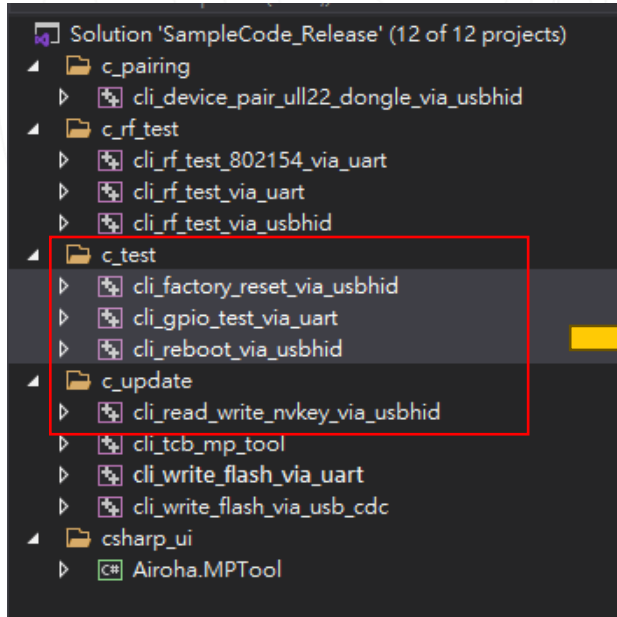
check UNIAA rule sample code

```
cli_device_pair_ul_gle_via_usbhid.cpp  X
cli_device_pair_ul22_dongle_via_usbhid  (Global Scope)

771 bool checkUniaaRule(uint8_t* _array)
772 {
773     // Check if the array starts with 0xFF or 0xFE
774     if (_array[0] == 0xFF || _array[0] == 0xFE)
775     {
776         return false;
777     }
778     // Check if the array is all 0x00 or all 0x01
779     bool all_zero = true;
780     bool all_one = true;
781     for (int i = 0; i < UNIAA_LEN; i++)
782     {
783         if (_array[i] != 0x00)
784         {
785             all_zero = false;
786         }
787         if (_array[i] != 0x01)
788         {
789             all_one = false;
790         }
791     }
792     if (all_zero || all_one)
793     {
794         return false;
795     }
796 }
```

1620 MP Library sample code

□ Sample code list



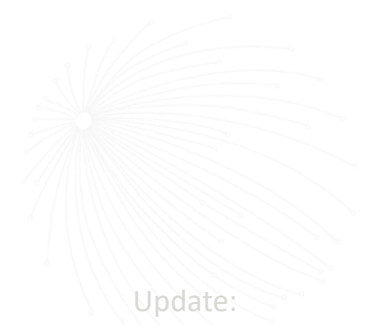
Sample code	Description
Factory Reset	Clear pairing info (exclude pre-pairing) Disable UART log
GPIO Test	Set GPIO high or low
Reboot DUT	APP WDT reset
Read/Write NV Key	For example : SN



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1620 DFU/FOTA



Update:

1620 DFU/FOTA

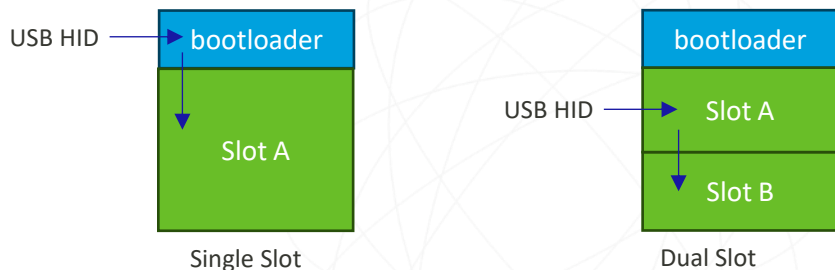
- ❑ **BT Data (HID) provide :**
 - Executable tool 、 DLL 、 full sample code 、 RACE command document.

Support project	Tool	SDK support	Upgrade file
BT Data DFU (USB-HID)	Airoha_BT_DFU	Full sample code	zephyr.signed.confirmed.bin
	Airoha_Bootloader_DFU	Full sample code	flash download.cfg
Smart Home FOTA (BLE)	Airoha_BT_Android_SDK	Sample code + Library	FOTA package
	Airoha_BT_iOS_SDK	Sample code + Library	FOTA package

1620 DFU sample code

- ❑ We provide 2 DFU solutions via USB-HID (Bootloader & Main Bin)
- ❑ C++ sample code support Windows/macOS

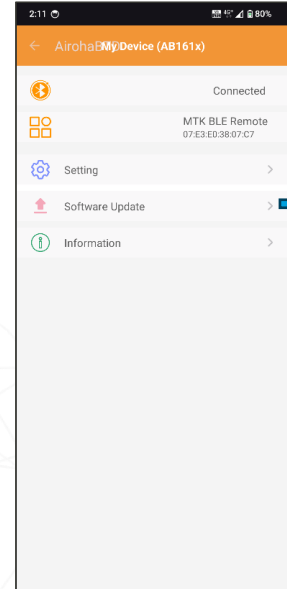
Tool	Flash layout	Data Transfer	Function when upgrading
Bootloader DFU	Single Slot	Bootloader	Disable
USB HID DFU	Dual Slot	Main bin	Enable



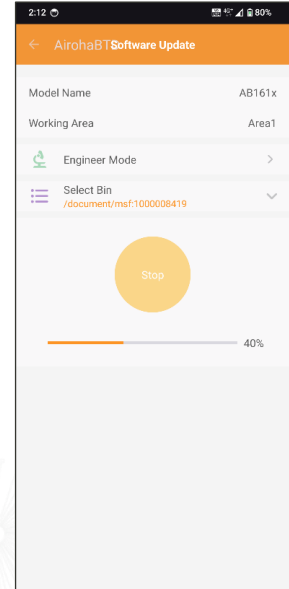
1620 FOTA sample code

- ❑ Phone APP support **Android & iOS**
- ❑ FOTA via **BLE**
- ❑ FOTA package tool
 - BIN file support compression & encryption

Function Menu



FOTA





1620 HID - Windows UT App

Update:

1620 HID Windows UT App

- ❑ Mouse上位機功能
- ❑ Using Airoha RACE command protocol

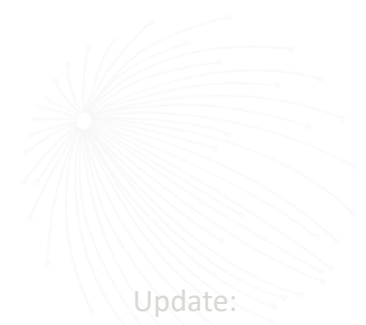
Tool	Function	Development method
Airoha_Windows_BT_D_UT_APP	Connection & Pairing	1. Using DLL & API 2. Using RACE command
	Report Rate (8K/4K/2K..) Profile Debounce setting DPI setting Sensor setting Key remap setting	
	Realtime RGB	



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Tool SDK Debug

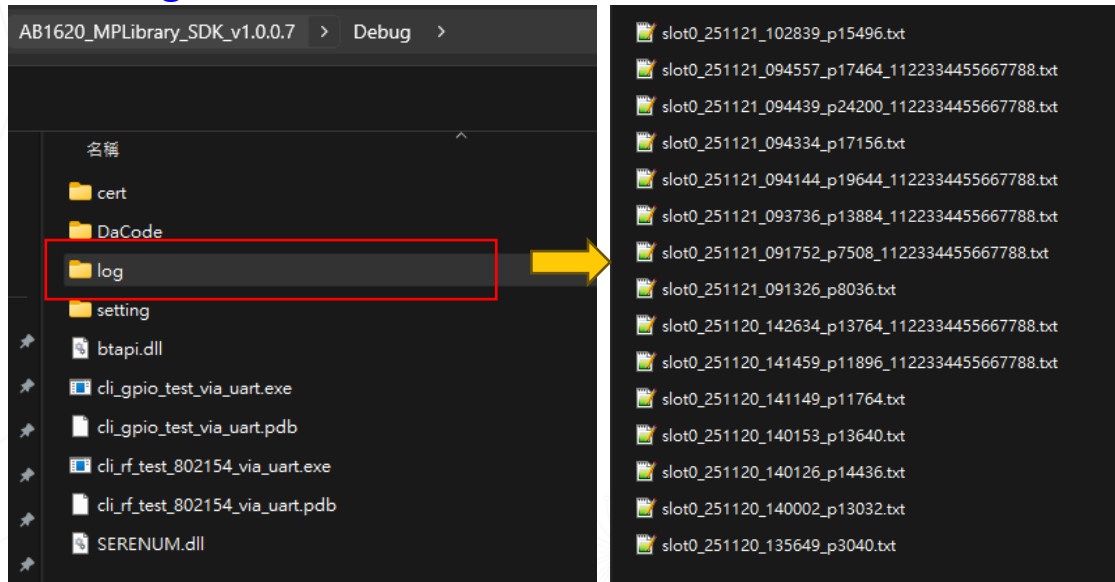


Update:

Tool SDK Debug

- ❑ Need **Tool log** & **FW log**
- ❑ Tool log
 - “log” folder
- ❑ FW log
 - Using AB1620 Logging tool
- ❑ Provide log via **JIRA (e-Service)**

Tool log





Thank you

Questions and Discussions

AIROHA

The Airoha logo is displayed in a bold, dark blue, sans-serif font. The letter 'H' is stylized with a horizontal line passing through its center. The logo is positioned on the left side of the slide, partially overlapping a large, faint, light blue circular graphic that resembles a fingerprint or a series of concentric, irregular rings.

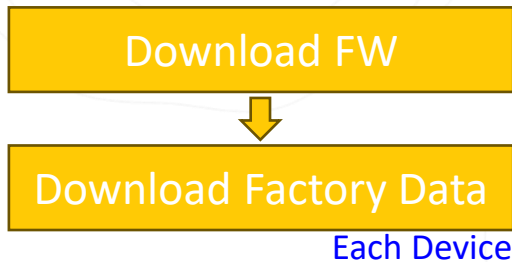
1620 Smart Home - Matter Factory Data

Update:

1620 Smart Home : Matter Factory Data

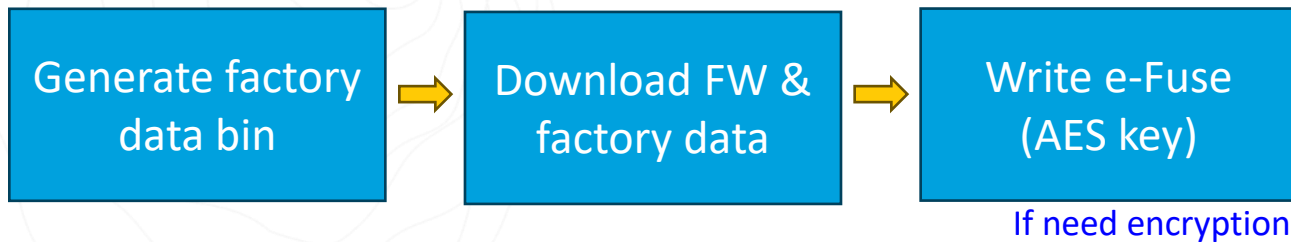
□ Each Matter device need unique data

- DAC / Serial number / Unique ID
- Passcode/ Spake2 salt / Discriminator



Category	Item	Example
Commissionable	Discriminator	0xF00
	Spake2 iterator	1000
	Spake2 salt	U1BBS0UyUCBLZXkgU2FsdA==
	passcode	20202021
Device Attestation	DAC cert	Matter-Development-DAC-FFF1-8006-Cert.der
	DAC key	Matter-Development-DAC-FFF1-8006-Key.der
	PAI cert	Matter-Development-PAI-FFF1-noPID-Cert.der
	Certification Declaration	Chip-Test-CD-FFF1.der
Device Instance	Vendor ID	65521
	Product ID	32774
	Vendor name	Airoha
	Product name	bulb
	Hardware version	1
	Hardware version string	prerelase
	Serial number	11223344556677889900
	Date	2025-04-11
	Unique ID	3ece0a408f13074bbcb657d73f135addc

1620 Smart Home : Matter Factory Data



Tool	Function
Airoha_Factory_Data_Tool	1. Generate Factory Data Bin
Airoha_Flash_Tool	1. Download Factory Data Bin 2. Write e-Fuse AES key
AB1620_MPLibrary_SDK	1. Download Factory Data Bin

Business Unit & Product doodles (spec)



Bluetooth



GNSS



Broadband



Ethernet

